**THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS**

A Complete First-Principles Reconstruction of Every Equation

# VOLUME VII

**(final volume)**

The Effective-Field Method and the Experiment:
Equations (B-5)–(B-10), the Numerical Procedure,
and the Comparison with Measurement

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**Where Volume VI ended.** Volume VI completed Shliomis & Morozov: the finite-amplitude equations (SM-35)–(SM-43), the square-wave solution, the viscosity map, and the conclusion that a negative *total* viscosity would require $\xi_c = 126$ — a field at which the rigid-dipole assumption itself fails. All 43 equations of that paper have now been derived from first principles.

Volume VII completes the project. It derives the effective-field method that Bacri *et al.* use in place of the phenomenological magnetisation equation, obtaining (B-5) through (B-9) from the Smoluchowski equation of Volume II; it shows how their numerical procedure works; and it goes through the experiment itself — the apparatus, the measured numbers, and the comparison between the theoretical and experimental isolines. Chapter numbers resume at 60, page numbers at 208.

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# Part XII

# The Experimental Paper and the Effective-Field Method

# Chapter 60

# Orientation: What the Experimental Paper Adds

## 60.1 The state of the account

Of the ten numbered equations of Bacri, Perzynski, Shliomis and Burde, five have already been derived in earlier volumes, because they are the same equations as Shliomis & Morozov's written in different notation:

| Equation | Statement | Equivalent to | Derived in |
|---|---|---|---|
| (B-1) | $6\eta\varphi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \boldsymbol{M}\times\boldsymbol{H}$ | (SM-11) | Sec. 14.2, Ch. 24 |
| (B-2) | $\Delta\eta = \tfrac{3}{2}\eta\varphi(\Omega-\omega_p)/\Omega$ | (SM-34) | Sec. 14.4, Ch. 48 |
| (B-3) | momentum equation with $\tfrac{1}{2}\operatorname{curl}(\boldsymbol{M}\times\boldsymbol{H})$ | (SM-12) | Sec. 14.3, Ch. 25 |
| (B-4) | $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{\tau_B}(\boldsymbol{M}-\boldsymbol{M}_0) - \frac{1}{6\eta\varphi}\boldsymbol{M}\times(\boldsymbol{M}\times\boldsymbol{H})$ | (SM-13) | Sec. 13.3, Ch. 26 |
| (B-10) | the grouping of the magnetic term with the viscous one | (SM-27) step | Sec. 14.4 |

Note the notational correspondences, fixed once and for all: their $\boldsymbol{\omega}_p$ is our $\boldsymbol{\Theta}$; their $\varphi M_s L(\xi)$ is our $nmL(\xi)$; their $\xi_0$ is the amplitude-based Langevin parameter, which is the $\xi$ of Sec. III of Shliomis & Morozov (Sec. 50.2).

What remains, and what this volume derives, is Eqs. (B-5)–(B-9): the *effective-field method*, which replaces the phenomenological magnetisation equation by a better-controlled closure of the Fokker–Planck hierarchy.

## 60.2 Why a better closure is needed

Section 13.4 of Volume II identified precisely where the phenomenological equation (SM-7) becomes an approximation: the exact first-moment equation contains the second moment $\langle e_i e_j\rangle$, and (SM-7) follows only after replacing it by its isotropic value $\tfrac{1}{3}\delta_{ij}$ — legitimate for $\xi \ll 1$ and no other regime.

Bacri *et al.* state the consequence they encountered. Their experimental data agree with the Shliomis–Morozov theory on the whole, but with substantial discrepancies in detail. Specifically, the theoretical $\Delta\eta(\xi)$ of Fig. 4 of Shliomis & Morozov has a maximum at $\omega\tau_B < 1$ and no minimum at $\omega\tau_B > 1$, whereas the measurements show the inverse situation. Their diagnosis

is exactly the one made in Sec. 13.4: the phenomenological equation works well for a stationary field of any intensity (whence the good agreement between McTague's measurements and Shliomis's 1972 theory), and for an alternating field of small amplitude *and* small frequency — but for finite $\xi$ and finite $\omega\tau_B$ together, a more rigorous magnetisation equation is required.

That equation was derived by Martsenyuk, Raikher and Shliomis from the Fokker–Planck equation for rotary diffusion, with the assistance of the effective-field method of Raikher and Shliomis. We now derive it.

## 60.3 Notation for this volume

| Symbol | Meaning | Type | Units |
|---|---|---|---|
| $\boldsymbol{\xi}$ | the *true* dimensionless field, $\boldsymbol{\xi} = m\boldsymbol{H}/kT$ | vector | — |
| $\boldsymbol{\xi}_e$ | the *effective* field: the fictitious static field whose equilibrium distribution equals the actual one | vector | — |
| $\xi_e = \lvert\boldsymbol{\xi}_e\rvert$ | its magnitude | scalar | — |
| $\boldsymbol{n} = \boldsymbol{\xi}_e/\xi_e$ | unit vector along the effective field | vector | — |
| $L_e = L(\xi_e)$ | Langevin function of the effective field | scalar | — |
| $L_3$ | the coefficient appearing in (B-6); shown in Sec. 62.6 to equal $L(\xi_e)$ | scalar | — |
| $\xi_0 = mH_0/kT$ | amplitude-based Langevin parameter | scalar | — |
| $F(\xi_e)$ | the function of (B-8) governing the transverse response | function | — |
| $g$ | normalised rotational viscosity, $g = \Delta\eta/\Delta\eta(\infty, 0)$ | scalar | — |
| $\eta_r(H, f)$ | measured reduced viscosity $[\eta(H, f) - \eta(0, 0)]/\eta(0, 0)$ | scalar | — |

# Chapter 61

# Equation (B-5): The Effective-Field Ansatz

$$\boldsymbol{M} = \varphi M_s \, L(\xi_e) \, \frac{\boldsymbol{\xi}_e}{\xi_e}. \tag{B-5}$$

## 61.1 Context

The paper introduces the method in one sentence: according to it, the nonequilibrium actual magnetisation is considered to be the equilibrium one, in the effective field $\boldsymbol{\xi}_e$. Equation (B-5) is that statement.

The idea deserves to be stated carefully, because it is a genuine physical hypothesis and not a mathematical identity.

**Assumption 61.1.1** (Effective-field closure)**.** At every instant, the orientational distribution function has the *form* of an equilibrium Boltzmann distribution, but in a fictitious field $\boldsymbol{\xi}_e(t)$ that is not the applied field:

$$W(\boldsymbol{e}, t) = \frac{\xi_e}{4\pi \sinh \xi_e} \, \exp\big(\boldsymbol{\xi}_e \cdot \boldsymbol{e}\big). \tag{61.1.1}$$

The dynamics is then carried entirely by the three components of $\boldsymbol{\xi}_e(t)$.

## 61.2 Why this is a good closure

Recall the difficulty. The exact moment equation, derived as Eq. (13.2.1) and its consequences in Volume II, reads

$$\frac{\mathrm{d}\langle e_i \rangle}{\mathrm{d}t} = \big(\boldsymbol{\Omega} \times \langle \boldsymbol{e} \rangle\big)_i + \frac{1}{2\tau_B}\Big[\xi_i - \big\langle e_i e_j \big\rangle \xi_j\Big] - \frac{\langle e_i \rangle}{\tau_B}. \tag{61.2.1}$$

*This equation is exact.* No approximation whatever has been made in reaching it: it follows from the Smoluchowski equation (11.6.1) by multiplying by $e_i$ and integrating, using Green's identity and the eigenvalue $\nabla_s^2 \boldsymbol{e} = -2\boldsymbol{e}$. Its only defect is that it is not closed: the first moment is driven by the second.

There are then three levels of treatment:

1. **Isotropic closure**, $\langle e_i e_j \rangle = \frac{1}{3}\delta_{ij}$: gives the phenomenological equation (SM-7). Exact only as $\xi \to 0$.

2. **Effective-field closure**, Assumption 61.1.1: expresses $\langle e_i e_j \rangle$ through $\langle \boldsymbol{e} \rangle$ by assuming the distribution retains its equilibrium *shape*. Exact in two limits — $\xi \to 0$, and any stationary field — and interpolating smoothly between them.

3. **Full Fokker–Planck**: solve the Smoluchowski equation (11.6.1) numerically for $W$ itself. Exact, expensive, and not what either paper does.

The effective-field method is level 2. Its merit is that it costs three unknowns instead of one and is exact wherever the answer is known independently.

*Remark* 61.2.1 (When is the ansatz exact?). In a *stationary* field with no flow, the true distribution *is* Boltzmann in the true field, so $\boldsymbol{\xi}_e = \boldsymbol{\xi}$ and (61.1.1) is exact. In a weak field of any time dependence, the distribution is $W \simeq (1 + \boldsymbol{a} \cdot \boldsymbol{e})/4\pi$ to first order in the small anisotropy, and (61.1.1) reproduces exactly that form with $\boldsymbol{\xi}_e = \boldsymbol{a}$. The ansatz therefore has no error at either end of the range, and the question is only how much it errs in between — which experiment, in the end, answers.

## 61.3 Derivation of (B-5) from the Ansatz

Given (61.1.1), the magnetisation follows from $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$, Eq. (4.5.1). The distribution (61.1.1) is identical in form to the equilibrium distribution (9.5.7) with $\xi \to \xi_e$ and the polar axis along $\boldsymbol{n} = \boldsymbol{\xi}_e/\xi_e$. Therefore the calculation of Chapter 10 applies verbatim, and Eq. (10.2.16) gives

$$\langle \boldsymbol{e} \rangle = L(\xi_e)\, \boldsymbol{n} = L(\xi_e) \frac{\boldsymbol{\xi}_e}{\xi_e}. \tag{61.3.1}$$

Multiplying by $nm = \varphi M_s$ [Sec. 10.4]:

$$\boldsymbol{M} = \varphi M_s L(\xi_e) \frac{\boldsymbol{\xi}_e}{\xi_e}, \tag{B-5}$$

■

## 61.4 The Second Moment

The whole point of the closure is that it determines $\langle e_i e_j \rangle$. We compute it now, since (B-6) cannot be derived without it.

**Result 61.4.1.** *For the distribution* (61.1.1),

$$\langle e_i e_j \rangle = \frac{L_e}{\xi_e}\, \delta_{ij} + \left(1 - \frac{3L_e}{\xi_e}\right) n_i n_j. \tag{61.4.1}$$

**Step 1: the tensor structure.** The distribution depends on $\boldsymbol{e}$ only through $\boldsymbol{e} \cdot \boldsymbol{n}$. The only tensors available to build $\langle e_i e_j \rangle$ are therefore $\delta_{ij}$ and $n_i n_j$, and it must be symmetric, so

$$\langle e_i e_j \rangle = A\, \delta_{ij} + B\, n_i n_j \tag{61.4.2}$$

for two scalars $A, B$ depending on $\xi_e$ alone.

**Step 2: first condition — the trace.** Contract $i = j$ in (61.4.2), using $\delta_{ii} = 3$, $n_i n_i = 1$ and $\langle e_i e_i \rangle = \langle 1 \rangle = 1$:

$$1 = 3A + B. \tag{61.4.3}$$

**Step 3: second condition — the longitudinal component.** Contract (61.4.2) with $n_i n_j$:

$$\big\langle (\boldsymbol{e}\cdot\boldsymbol{n})^2 \big\rangle = A + B. \tag{61.4.4}$$

So we need $\langle u^2\rangle$ with $u = \boldsymbol{e}\cdot\boldsymbol{n} = \cos\theta$.

**Step 4: compute $\langle u^2\rangle$.** Write the partition function of Eq. (9.5.8), $Z(\xi_e) = \int e^{\xi_e u}\mathrm{d}\Omega_e$, and recall from Chapter 10 that

$$\langle u\rangle = \frac{Z'}{Z} = L(\xi_e), \tag{61.4.5}$$

where the prime is $\mathrm{d}/\mathrm{d}\xi_e$, because differentiating $e^{\xi_e u}$ with respect to the parameter brings down a factor $u$. Differentiating again,

$$\frac{\mathrm{d}\langle u\rangle}{\mathrm{d}\xi_e} = \frac{\mathrm{d}}{\mathrm{d}\xi_e}\frac{Z'}{Z} = \frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2 = \big\langle u^2\big\rangle - \langle u\rangle^2, \tag{61.4.6}$$

so

$$\big\langle u^2\big\rangle = L'(\xi_e) + L^2(\xi_e). \tag{61.4.7}$$

Now differentiate $L = \coth\xi - 1/\xi$:

$$L' = -\frac{1}{\sinh^2\xi} + \frac{1}{\xi^2}, \tag{61.4.8}$$

using $\mathrm{d}(\coth\xi)/\mathrm{d}\xi = -1/\sinh^2\xi$ (from the quotient rule applied to $\cosh/\sinh$ together with $\cosh^2 - \sinh^2 = 1$). And

$$L^2 = \left(\coth\xi - \frac{1}{\xi}\right)^2 = \coth^2\xi - \frac{2\coth\xi}{\xi} + \frac{1}{\xi^2} \tag{61.4.9}$$

$$= 1 + \frac{1}{\sinh^2\xi} - \frac{2\coth\xi}{\xi} + \frac{1}{\xi^2}, \tag{61.4.10}$$

where $\coth^2 = 1 + 1/\sinh^2$ (divide $\cosh^2 - \sinh^2 = 1$ by $\sinh^2$). Adding, the $1/\sinh^2$ terms cancel:

$$\big\langle u^2\big\rangle = \frac{1}{\xi^2} - \frac{1}{\sinh^2\xi} + 1 + \frac{1}{\sinh^2\xi} - \frac{2\coth\xi}{\xi} + \frac{1}{\xi^2} \tag{61.4.11}$$

$$= 1 + \frac{2}{\xi^2} - \frac{2\coth\xi}{\xi} \tag{61.4.12}$$

$$= 1 - \frac{2}{\xi}\left(\coth\xi - \frac{1}{\xi}\right) \tag{61.4.13}$$

$$= 1 - \frac{2L(\xi)}{\xi}. \tag{61.4.14}$$

**Step 5: solve for $A$ and $B$.** From (61.4.4) and (61.4.14), $A + B = 1 - 2L_e/\xi_e$. Subtracting this from (61.4.3):

$$3A + B - \big(A + B\big) = 1 - \left(1 - \frac{2L_e}{\xi_e}\right) \quad\Longrightarrow\quad 2A = \frac{2L_e}{\xi_e} \quad\Longrightarrow\quad A = \frac{L_e}{\xi_e}. \tag{61.4.15}$$

Then from (61.4.3), $B = 1 - 3A = 1 - 3L_e/\xi_e$. Substituting into (61.4.2) gives (61.4.1). ■

**Sanity checks.** At $\xi_e \to 0$, $L_e \to \xi_e/3$, so $A \to 1/3$ and $B \to 1 - 1 = 0$: the isotropic result $\frac{1}{3}\delta_{ij}$ of Eq. (13.2.9). ✓ At $\xi_e \to \infty$, $L_e \to 1$, so $A \to 0$ and $B \to 1$: all moments aligned, $\langle e_i e_j \rangle = n_i n_j$. ✓

## 61.5 Connection to the Next Equation

Inserting (61.3.1) and (61.4.1) into the exact moment equation (61.2.1) closes the system and produces (B-6).

# Chapter 62

# Equation (B-6): The Effective-Field Evolution Equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\right)=\boldsymbol{\Omega}\times\left(L_e\frac{\boldsymbol{\xi}_e}{\xi_e}\right)-\frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e-\boldsymbol{\xi})-\frac{1}{2\tau_B\xi_e^2}\left(1-\frac{3L_3}{\xi_e}\right)\boldsymbol{\xi}_e\times(\boldsymbol{\xi}_e\times\boldsymbol{\xi}). \tag{B-6}$$

## 62.1 Context

This is the magnetisation equation of Martsenyuk, Raikher and Shliomis, written for the effective field rather than for $\boldsymbol{M}$. It replaces (SM-7)/(B-4), and it is the equation Bacri *et al.* actually integrate.

## 62.2 Assumptions

1. The Smoluchowski equation (11.6.1) — itself derived from probability conservation, inertialess torque balance and the Einstein relation.
2. The effective-field closure, Assumption 61.1.1.
3. Rigid dipoles, monodisperse, non-interacting: as always.

Note what is *not* assumed: nothing about the size of $\xi$, nothing about the frequency, nothing about $\Omega\tau_B$. Equation (B-6) is a closed nonlinear equation valid, within its closure, for arbitrary fields and flows.

## 62.3 Mathematical Background: the exact moment equation

We restate it because everything follows from it. From Volume II, Sec. 13.2, the three contributions to the first moment were the flow term (13.2.4), the diffusion term (13.2.6), and the magnetic term (13.2.8), before any closure. Assembling them with $D_r = 1/2\tau_B$ [Eq. (11.5.3)] and $m\boldsymbol{H}/kT = \boldsymbol{\xi}$:

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t}=\big(\boldsymbol{\Omega}\times\langle\boldsymbol{e}\rangle\big)_i+\frac{1}{2\tau_B}\Big[\xi_i-\big\langle e_ie_j\big\rangle\xi_j\Big]-\frac{\langle e_i\rangle}{\tau_B}. \tag{62.3.1}$$

*Check of the coefficient:* the magnetic term in Volume II was $(m/\zeta_r)[H_i - \langle e_ie_j\rangle H_j]$. Using the Einstein relation $D_r = kT/\zeta_r$, $m/\zeta_r = mD_r/kT$, and $mH_i/kT = \xi_i$, this is $D_r[\xi_i - \langle e_ie_j\rangle\xi_j] = \frac{1}{2\tau_B}[\cdots]$. ✓

## 62.4 Step-by-Step Derivation

**Step 1: insert the closure.** Substitute (61.3.1) and (61.4.1) into (62.3.1). The second moment contracted with the field is

$$\langle e_i e_j\rangle \xi_j = \frac{L_e}{\xi_e}\delta_{ij}\xi_j + \left(1 - \frac{3L_e}{\xi_e}\right) n_i n_j \xi_j \tag{62.4.1}$$

$$= \frac{L_e}{\xi_e}\xi_i + \left(1 - \frac{3L_e}{\xi_e}\right) n_i\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big). \tag{62.4.2}$$

Hence, in vector form,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\boldsymbol{n}\big) = \boldsymbol{\Omega}\times\big(L_e\boldsymbol{n}\big) + \frac{1}{2\tau_B}\left[\boldsymbol{\xi} - \frac{L_e}{\xi_e}\boldsymbol{\xi} - \left(1 - \frac{3L_e}{\xi_e}\right)\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)\right] - \frac{L_e\boldsymbol{n}}{\tau_B}. \tag{62.4.3}$$

This is already a closed equation for $\boldsymbol{\xi}_e$. What remains is to rearrange it into the form printed by the authors.

**Step 2: identify the target form.** Expand the two magnetic terms of (B-6), using $\boldsymbol{\xi}_e = \xi_e\boldsymbol{n}$ and the triple product (1.7.1):

$$\boldsymbol{\xi}_e\times\big(\boldsymbol{\xi}_e\times\boldsymbol{\xi}\big) = \boldsymbol{\xi}_e\big(\boldsymbol{\xi}_e\cdot\boldsymbol{\xi}\big) - \boldsymbol{\xi}\big(\boldsymbol{\xi}_e\cdot\boldsymbol{\xi}_e\big) \tag{62.4.4}$$

$$= \xi_e^2\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big) - \boldsymbol{\xi}\Big]. \tag{62.4.5}$$

Therefore the last term of (B-6) is

$$-\frac{1}{2\tau_B\xi_e^2}\left(1 - \frac{3L_3}{\xi_e}\right)\xi_e^2\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big) - \boldsymbol{\xi}\Big] = -\frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right)\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big) - \boldsymbol{\xi}\Big], \tag{62.4.6}$$

the factors $\xi_e^2$ cancelling. And the middle term of (B-6) is

$$-\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\xi_e\boldsymbol{n} - \boldsymbol{\xi}\big) = -\frac{L_e\boldsymbol{n}}{\tau_B} + \frac{L_e}{\tau_B\xi_e}\boldsymbol{\xi}. \tag{62.4.7}$$

So the right-hand side of (B-6), apart from the flow term, is

$$-\frac{L_e\boldsymbol{n}}{\tau_B} + \left[\frac{L_e}{\tau_B\xi_e} + \frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right)\right]\boldsymbol{\xi} - \frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right)\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big). \tag{62.4.8}$$

**Step 3: compare with the derived equation.** Rewriting (62.4.3) in the same order:

$$-\frac{L_e\boldsymbol{n}}{\tau_B} + \frac{1}{2\tau_B}\left[1 - \frac{L_e}{\xi_e}\right]\boldsymbol{\xi} - \frac{1}{2\tau_B}\left(1 - \frac{3L_e}{\xi_e}\right)\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big). \tag{62.4.9}$$

**Step 4: match, term by term.** The first terms are identical. Matching the coefficients of $\boldsymbol{n}(\boldsymbol{n}\cdot\boldsymbol{\xi})$ in (62.4.8) and (62.4.9) requires

$$L_3 = L_e = L\big(\xi_e\big). \tag{62.4.10}$$

Matching the coefficients of $\boldsymbol{\xi}$ requires

$$\frac{L_e}{\tau_B\xi_e} + \frac{1}{2\tau_B}\left(1 - \frac{3L_3}{\xi_e}\right) = \frac{1}{2\tau_B}\left(1 - \frac{L_e}{\xi_e}\right), \tag{62.4.11}$$

i.e., multiplying by $2\tau_B$,

$$\frac{2L_e}{\xi_e} + 1 - \frac{3L_3}{\xi_e} = 1 - \frac{L_e}{\xi_e} \quad\Longleftrightarrow\quad \frac{3L_e}{\xi_e} = \frac{3L_3}{\xi_e} \quad\Longleftrightarrow\quad L_3 = L_e, \tag{62.4.12}$$

the *same* condition. With (62.4.10) the two expressions agree identically, and (B-6) is established. ■

## 62.5 The Identification of $L_3$

The symbol $L_3$ in the printed equation is not defined in the letter, which cites the original derivation for it. The matching above determines it unambiguously: *within the effective-field closure,* $L_3 = L(\xi_e)$. Two independent confirmations follow:

1. Chapter 64 derives (B-8) from (B-6). The derivation reproduces the printed (B-8) exactly if and only if $L_3 = L_e$; any other value leaves an uncancelled residue. This is a genuinely independent check, since (B-8) is a different equation in the same paper.
2. The physical origin of the term is the second moment (61.4.1), whose anisotropic coefficient is $1 - 3L_e/\xi_e$ — exactly the combination appearing in (B-6) with $L_3$ in place of $L_e$. The subscript 3 evidently labels this coefficient as arising from the second-rank (three-component) part of the orientational distribution.

## 62.6 Physical Interpretation

The three terms on the right of (B-6) have distinct meanings.

- $\boldsymbol{\Omega} \times (L_e \boldsymbol{n})$: the magnetisation is carried around by the flow, as in every version of the theory.
- $-\frac{1}{\tau_B}\frac{L_e}{\xi_e}(\boldsymbol{\xi}_e - \boldsymbol{\xi})$: *longitudinal* relaxation. It drives the effective field towards the true field. If the true and effective fields were parallel this would be the only magnetic term, and (B-6) would reduce to a scalar relaxation equation — which is exactly what happens in Chapter 63.
- $-\frac{1}{2\tau_B \xi_e^2}(1 - 3L_e/\xi_e)\,\boldsymbol{\xi}_e \times (\boldsymbol{\xi}_e \times \boldsymbol{\xi})$: *transverse* relaxation, active only when $\boldsymbol{\xi}_e$ and $\boldsymbol{\xi}$ are not parallel. By (62.4.5) it is proportional to the component of $\boldsymbol{\xi}$ perpendicular to $\boldsymbol{\xi}_e$.

**Anisotropic relaxation.** The essential improvement over (SM-7) is that these two relaxation channels now have *different rates*. Reading off from (62.4.7) and (62.4.6), and dividing by $L_e$ to convert to rates for $\boldsymbol{n}$:

$$\frac{1}{\tau_{\parallel}} \sim \frac{1}{\tau_B}\frac{L_e}{\xi_e} \cdot \frac{1}{\mathrm{d}L_e/\mathrm{d}\xi_e}, \qquad \frac{1}{\tau_{\perp}} \sim \frac{1}{2\tau_B}\left(1 - \frac{3L_e}{\xi_e}\right)\frac{\xi_e}{L_e}. \tag{62.6.1}$$

At $\xi_e \to 0$ both reduce to $1/\tau_B$ and the theory collapses back onto (SM-7), as it must. At large $\xi_e$ they differ by a factor of order $\xi_e$: a strongly magnetised ferrofluid relaxes its magnetisation *direction* at a completely different rate from its *magnitude*. The phenomenological equation, which uses one time constant for both, cannot capture this — and it is precisely in the regime of large $\xi$ and large $\omega\tau_B$, where both channels are active, that Bacri *et al.* found it failing.

## 62.7 Connection to the Next Equation

To use (B-6) we expand in the small parameter $\Omega\tau_B$. The zeroth order gives (B-7).

# Chapter 63

# Equation (B-7): The Zero Approximation

$$\tau_B \frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}\left(1 - \frac{\xi_0}{\xi_e}\cos\omega t\right). \tag{B-7}$$

## 63.1 Context

The paper explains the ordering: the problem contains a small dimensionless parameter $\Omega\tau_B$. A transient birefringence experiment gives $\tau_B \simeq 1.6$ ms for this fluid, so $\Omega\tau_B \ll 1$ is satisfied for any vorticity below 60 $\mathrm{s}^{-1}$, which is always the case in their apparatus. For small $\Omega\tau_B$ the solution of (B-3) and (B-6) can be found by perturbation theory, and (B-7) is the zeroth-order equation.

**Verifying the stated bound.** $\Omega\tau_B \ll 1$ with $\tau_B = 1.6 \times 10^{-3}$ s means $\Omega \ll 625$ $\mathrm{s}^{-1}$; the paper's statement that the inequality holds for $\Omega$ smaller than 60 $\mathrm{s}^{-1}$ corresponds to $\Omega\tau_B \lesssim 0.1$, a comfortable margin. ✓

## 63.2 Assumptions

1. $\Omega = 0$ at this order.
2. The applied field is $\boldsymbol{H} = (H_0\cos\omega t, 0, 0)$, so $\boldsymbol{\xi} = \xi_0\cos\omega t\,\hat{\boldsymbol{x}}$ with $\xi_0 = mH_0/kT$.
3. With no flow, the effective field has no reason to leave the field axis: $\boldsymbol{\xi}_e = \xi_e(t)\hat{\boldsymbol{x}}$, with $\xi_e$ allowed to be of either sign.

## 63.3 Step-by-Step Derivation

**Step 1: the transverse term vanishes.** With $\boldsymbol{\xi}_e \parallel \boldsymbol{\xi}$, the cross product $\boldsymbol{\xi}_e \times \boldsymbol{\xi} = \mathbf{0}$ by Eq. (1.5.6), so the last term of (B-6) is identically zero. So is the flow term, since $\boldsymbol{\Omega} = \mathbf{0}$.

**Step 2: what remains.** With $\boldsymbol{n} = \hat{\boldsymbol{x}}$ constant,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\hat{\boldsymbol{x}}\big) = -\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\xi_e - \xi_0\cos\omega t\big)\hat{\boldsymbol{x}}, \tag{63.3.1}$$

i.e., as a scalar equation,

$$\frac{\mathrm{d}L_e}{\mathrm{d}t} = -\frac{L_e}{\tau_B \xi_e}\big(\xi_e - \xi_0 \cos\omega t\big). \tag{63.3.2}$$

**Step 3: change the dependent variable.** We want an equation for $\xi_e$, not for $L_e$. By the chain rule,

$$\frac{\mathrm{d}L_e}{\mathrm{d}t} = \frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}\frac{\mathrm{d}\xi_e}{\mathrm{d}t}. \tag{63.3.3}$$

Substituting into (63.3.2) and solving for $\mathrm{d}\xi_e/\mathrm{d}t$:

$$\frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\frac{1}{\mathrm{d}L_e/\mathrm{d}\xi_e} \cdot \frac{L_e}{\tau_B \xi_e}\big(\xi_e - \xi_0 \cos\omega t\big). \tag{63.3.4}$$

**Step 4: recognise the logarithmic derivative.** Since

$$\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e} = \frac{1}{L_e}\frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}, \qquad \text{so} \qquad \frac{L_e}{\mathrm{d}L_e/\mathrm{d}\xi_e} = \left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}, \tag{63.3.5}$$

Eq. (63.3.4) becomes

$$\frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1} \frac{1}{\tau_B \xi_e}\big(\xi_e - \xi_0 \cos\omega t\big). \tag{63.3.6}$$

**Step 5: tidy.** Multiply by $\tau_B$ and divide the bracket by $\xi_e$:

$$\tau_B \frac{\mathrm{d}\xi_e}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1} \left(1 - \frac{\xi_0}{\xi_e}\cos\omega t\right), \tag{B-7}$$

which is (B-7). ■

## 63.4 Physical Interpretation

**The fixed point.** The right-hand side vanishes when $\xi_e = \xi_0 \cos\omega t$, i.e. when the effective field equals the true field. That is the quasi-static state, reached whenever the field varies slowly compared with $\tau_B$. Everything interesting comes from the failure to reach it.

**The relaxation rate is field-dependent.** The prefactor $(\mathrm{d}\ln L_e/\mathrm{d}\xi_e)^{-1}$ is the new feature compared with (SM-7), where the corresponding factor was the constant 1. Its behaviour:

- Weak field: $L \simeq \xi/3$, so $\ln L \simeq \ln\xi - \ln 3$ and $\mathrm{d}\ln L/\mathrm{d}\xi = 1/\xi$, giving $(\mathrm{d}\ln L/\mathrm{d}\xi)^{-1} = \xi_e$. Then (B-7) reads $\tau_B \dot{\xi}_e = -(\xi_e - \xi_0\cos\omega t)$ — exactly the linear Debye relaxation of Volume V, Eq. (44.4.2). ✓
- Strong field: $L \to 1 - 1/\xi$, so $\mathrm{d}L/\mathrm{d}\xi \to 1/\xi^2$ and $\mathrm{d}\ln L/\mathrm{d}\xi \to 1/\xi^2$, giving $(\mathrm{d}\ln L/\mathrm{d}\xi)^{-1} \to \xi_e^2$. The restoring term is enormously enhanced: a strongly magnetised system is stiff, and its effective field tracks the applied one much more closely than the weak-field theory would suggest.

This stiffening is the single most important physical difference between the effective-field theory and the phenomenological one, and it is what shifts the computed curves into agreement with the measurements.

## 63.5 Connection to the Next Equation

At first order in $\Omega\tau_B$ the effective field acquires a transverse component, and its amplitude is governed by (B-8).

# Chapter 64

# Equation (B-8): The First-Order Response

$$\tau_B \frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \frac{1}{2}\left(\frac{1}{L_e} - \frac{1}{\xi_e}\right)\xi_0 F\cos\omega t. \tag{B-8}$$

## 64.1 Context

The paper states: in the linear approximation in $\Omega\tau_B$, Eq. (B-6) yields

$$\boldsymbol{M}^{(1)} = M^{(0)} F(\xi_e)\,\tau_B\,\boldsymbol{\Omega}\times\hat{\boldsymbol{h}}, \qquad M^{(0)} = \varphi M_s L(\xi_e), \tag{64.1.1}$$

and the function $F$ is determined by (B-8). This is the analogue of the function $G$ of Volume VI: it carries the transverse response, and it is the object that will be correlated with the field to give the torque.

## 64.2 Symbols

$\hat{\boldsymbol{h}} = \hat{\boldsymbol{x}}$ is the unit vector along the applied field, so $\boldsymbol{\Omega}\times\hat{\boldsymbol{h}} = \Omega\hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}} = \Omega\hat{\boldsymbol{y}}$: the first-order magnetisation is transverse, as it must be. $F$ is dimensionless.

## 64.3 Set-up

Write the effective field to first order in $\Omega\tau_B$ as

$$\boldsymbol{\xi}_e = \xi_e\hat{\boldsymbol{x}} + \xi_e^{(1)}\hat{\boldsymbol{y}}, \qquad \boldsymbol{n} = \hat{\boldsymbol{x}} + \varepsilon\,\hat{\boldsymbol{y}}, \qquad \varepsilon \equiv \frac{\xi_e^{(1)}}{\xi_e} = O\big(\Omega\tau_B\big). \tag{64.3.1}$$

The magnitude $|\boldsymbol{\xi}_e| = \xi_e\sqrt{1+\varepsilon^2} = \xi_e + O(\varepsilon^2)$ is unchanged at first order, so $L_e$ and every function of $\xi_e$ retain their zeroth-order values.

**Definition of $F$.** By (B-5), the transverse magnetisation is

$$M_y = \varphi M_s L_e\,\varepsilon = M^{(0)}\varepsilon, \tag{64.3.2}$$

and comparison with (64.1.1), whose $y$ component is $M^{(0)}F\tau_B\Omega$, gives

$$\boxed{\varepsilon = F\,\tau_B\,\Omega} \tag{64.3.3}$$

This *defines* $F$; the task is to find its equation of motion.

## 64.4 Step-by-Step Derivation

We take the $\hat{\boldsymbol{y}}$ component of (B-6), keeping terms of first order in $\varepsilon$.

**Step 1: the left-hand side.**

$$\left[\frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\boldsymbol{n}\big)\right]_y = \frac{\mathrm{d}}{\mathrm{d}t}\big(L_e\varepsilon\big) = \tau_B\Omega\,\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big), \tag{64.4.1}$$

using (64.3.3) and treating $\Omega$ and $\tau_B$ as constants.

**Step 2: the flow term.**

$$\Big[\boldsymbol{\Omega}\times\big(L_e\boldsymbol{n}\big)\Big]_y = \Big[\Omega\hat{\boldsymbol{z}}\times L_e\big(\hat{\boldsymbol{x}}+\varepsilon\hat{\boldsymbol{y}}\big)\Big]_y \tag{64.4.2}$$

$$= \Big[\Omega L_e\big(\hat{\boldsymbol{y}}-\varepsilon\hat{\boldsymbol{x}}\big)\Big]_y \qquad \text{using } \hat{\boldsymbol{z}}\times\hat{\boldsymbol{x}}=\hat{\boldsymbol{y}},\ \hat{\boldsymbol{z}}\times\hat{\boldsymbol{y}}=-\hat{\boldsymbol{x}} \tag{64.4.3}$$

$$= \Omega L_e. \tag{64.4.4}$$

Note this term is *zeroth* order in $\varepsilon$: it is the source that creates the transverse component in the first place.

**Step 3: the longitudinal relaxation term.** With $\boldsymbol{\xi}_e = \xi_e\hat{\boldsymbol{x}} + \xi_e\varepsilon\hat{\boldsymbol{y}}$ and $\boldsymbol{\xi} = \xi_0\cos\omega t\,\hat{\boldsymbol{x}}$ (no $y$ component),

$$\left[-\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\boldsymbol{\xi}_e-\boldsymbol{\xi}\big)\right]_y = -\frac{1}{\tau_B}\frac{L_e}{\xi_e}\big(\xi_e\varepsilon\big) = -\frac{L_e\varepsilon}{\tau_B}. \tag{64.4.5}$$

**Step 4: the transverse relaxation term.** From (62.4.5),

$$\boldsymbol{\xi}_e\times\big(\boldsymbol{\xi}_e\times\boldsymbol{\xi}\big) = \xi_e^2\Big[\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)-\boldsymbol{\xi}\Big]. \tag{64.4.6}$$

Compute the bracket:

$$\boldsymbol{n}\cdot\boldsymbol{\xi} = \big(\hat{\boldsymbol{x}}+\varepsilon\hat{\boldsymbol{y}}\big)\cdot\big(\xi_0\cos\omega t\,\hat{\boldsymbol{x}}\big) = \xi_0\cos\omega t, \tag{64.4.7}$$

$$\boldsymbol{n}\big(\boldsymbol{n}\cdot\boldsymbol{\xi}\big)-\boldsymbol{\xi} = \xi_0\cos\omega t\big(\hat{\boldsymbol{x}}+\varepsilon\hat{\boldsymbol{y}}\big)-\xi_0\cos\omega t\,\hat{\boldsymbol{x}} = \varepsilon\,\xi_0\cos\omega t\,\hat{\boldsymbol{y}}. \tag{64.4.8}$$

The $\hat{\boldsymbol{x}}$ components cancel exactly — as they must, since a vector triple product of this form is perpendicular to $\boldsymbol{\xi}_e$ — leaving a purely transverse result. Hence

$$\left[-\frac{1}{2\tau_B\xi_e^2}\left(1-\frac{3L_e}{\xi_e}\right)\boldsymbol{\xi}_e\times\big(\boldsymbol{\xi}_e\times\boldsymbol{\xi}\big)\right]_y = -\frac{1}{2\tau_B}\left(1-\frac{3L_e}{\xi_e}\right)\varepsilon\,\xi_0\cos\omega t, \tag{64.4.9}$$

where $L_3 = L_e$ from (62.4.10) has been used.

**Step 5: assemble.** Equating (64.4.1) to the sum of (64.4.4), (64.4.5) and (64.4.9), and substituting $\varepsilon = F\tau_B\Omega$:

$$\tau_B\Omega\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big) = \Omega L_e - \frac{L_e}{\tau_B}F\tau_B\Omega - \frac{1}{2\tau_B}\left(1-\frac{3L_e}{\xi_e}\right)F\tau_B\Omega\,\xi_0\cos\omega t. \tag{64.4.10}$$

Divide throughout by $\Omega$ (nonzero) and simplify each term:

$$\tau_B\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big) = L_e - L_eF - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t. \tag{64.4.11}$$

**Step 6: expand the left-hand side.** By the product rule,

$$\tau_B\frac{\mathrm{d}}{\mathrm{d}t}\big(L_eF\big) = \tau_BL_e\frac{\mathrm{d}F}{\mathrm{d}t} + \tau_BF\frac{\mathrm{d}L_e}{\mathrm{d}t}. \tag{64.4.12}$$

We must therefore know $\tau_B\,\mathrm{d}L_e/\mathrm{d}t$, which the zeroth-order equation supplies.

**Step 7: evaluate $\tau_B\,\mathrm{d}L_e/\mathrm{d}t$ from (B-7).** By the chain rule and (63.3.5),

$$\tau_B\frac{\mathrm{d}L_e}{\mathrm{d}t} = \frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}\cdot\tau_B\frac{\mathrm{d}\xi_e}{\mathrm{d}t} \tag{64.4.13}$$

$$= \frac{\mathrm{d}L_e}{\mathrm{d}\xi_e}\cdot\left[-\left(\frac{\mathrm{d}\ln L_e}{\mathrm{d}\xi_e}\right)^{-1}\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right)\right] \tag{64.4.14}$$

$$= -\frac{\mathrm{d}L_e/\mathrm{d}\xi_e}{\mathrm{d}\ln L_e/\mathrm{d}\xi_e}\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right) \tag{64.4.15}$$

$$= -L_e\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right), \tag{64.4.16}$$

the last step because $\mathrm{d}\ln L_e/\mathrm{d}\xi_e = (1/L_e)\,\mathrm{d}L_e/\mathrm{d}\xi_e$, so the ratio is simply $L_e$. *This clean cancellation is the reason the authors wrote (B-7) in terms of the logarithmic derivative.*

**Step 8: substitute and collect.** Insert (64.4.12) with (64.4.16) into (64.4.11):

$$\tau_BL_e\frac{\mathrm{d}F}{\mathrm{d}t} - L_eF\left(1-\frac{\xi_0}{\xi_e}\cos\omega t\right) = L_e - L_eF - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t. \tag{64.4.17}$$

Move the second term of the left side across:

$$\tau_BL_e\frac{\mathrm{d}F}{\mathrm{d}t} = L_e - L_eF - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t + L_eF - \frac{L_eF\xi_0\cos\omega t}{\xi_e} \tag{64.4.18}$$

$$= L_e - \frac{1}{2}\left(1-\frac{3L_e}{\xi_e}\right)\xi_0F\cos\omega t - \frac{L_e\xi_0F\cos\omega t}{\xi_e}, \tag{64.4.19}$$

the terms $\mp L_eF$ having cancelled.

**Step 9: divide by $L_e$.**

$$\tau_B\frac{\mathrm{d}F}{\mathrm{d}t} = 1 - \xi_0F\cos\omega t\left[\frac{1}{2L_e}\left(1-\frac{3L_e}{\xi_e}\right)+\frac{1}{\xi_e}\right] \tag{64.4.20}$$

$$= 1 - \xi_0F\cos\omega t\left[\frac{1}{2L_e}-\frac{3}{2\xi_e}+\frac{1}{\xi_e}\right] \tag{64.4.21}$$

$$= 1 - \xi_0 F \cos\omega t \left[ \frac{1}{2L_e} - \frac{1}{2\xi_e} \right] \tag{64.4.22}$$

$$= 1 - \frac{1}{2}\left( \frac{1}{L_e} - \frac{1}{\xi_e} \right) \xi_0 F \cos\omega t, \tag{B-8}$$

using $-3/2 + 1 = -1/2$ in the third line. This is (B-8) exactly. ■

*Remark* 64.4.1 (The check on $L_3$). Had we kept $L_3$ general in Step 4, the bracket in Step 9 would have been $\frac{1}{2L_e} - \frac{3L_3}{2L_e\xi_e} + \frac{1}{\xi_e}$, which equals the printed $\frac{1}{2L_e} - \frac{1}{2\xi_e}$ only if $L_3 = L_e$. Equation (B-8) as published therefore confirms the identification (62.4.10) independently of the matching argument of Sec. 62.5.

## 64.5 Consistency Check: the Weak-Field Limit

At $\xi_e \ll 1$, $L_e \to \xi_e/3$, so

$$\frac{1}{2}\left( \frac{1}{L_e} - \frac{1}{\xi_e} \right) \to \frac{1}{2}\left( \frac{3}{\xi_e} - \frac{1}{\xi_e} \right) = \frac{1}{\xi_e}, \tag{64.5.1}$$

and (B-8) becomes $\tau_B \dot{F} = 1 - (\xi_0/\xi_e) F \cos\omega t$. Meanwhile, in the same limit, the zeroth-order equation reads $\tau_B \dot{\xi}_e = -(\xi_e - \xi_0 \cos\omega t)$ (Sec. 63.5). The transverse magnetisation is $M_y = M^{(0)} F \tau_B \Omega$ with $M^{(0)} = \varphi M_s \xi_e / 3 = \chi H_e$ — i.e. exactly the structure of the weak-field theory of Volume V, in which $M_y$ was likewise proportional to $\Omega\tau_B$ times a function of frequency. ✓

## 64.6 Connection to the Next Equation

With $\xi_e(t)$ from (B-7) and $F(t)$ from (B-8), the averaged torque and hence the viscosity follow. That is (B-9).

# Chapter 65

# Equation (B-9): The Mean Torque and the Function $g$

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -6\eta\varphi\, g\,\boldsymbol{\Omega}, \qquad g = \frac{1}{2}\,\xi_0\,\overline{\cos\omega t\, L(\xi_e)F(\xi_e)}. \tag{B-9}$$

## 65.1 Context

The paper notes that the torque may be averaged over the period of field variation $2\pi/\omega$, because in their experiments this period is always much smaller than the characteristic hydrodynamic time $\rho d^2/\eta$. This is the same two-scale argument proved in Chapter 40; here we check the criterion with their numbers.

**Verifying the scale separation.** With $d = 0.1$ cm (the capillary diameter), $\rho \approx 2\ \mathrm{g\,cm^{-3}}$ and $\eta = 0.77$ P,

$$\frac{\rho d^2}{\eta} = \frac{2\times10^{-2}}{0.77} = 2.6\times10^{-2}\ \mathrm{s} = 26\ \mathrm{ms}, \tag{65.1.1}$$

against field periods from 19 ms (at 52 Hz) down to 0.68 ms (at 1480 Hz). The separation is marginal at the lowest frequency and excellent at the highest, which — as in Sec. 40.3 — is the regime where the effect matters. ✓

## 65.2 Step-by-Step Derivation

**Step 1: the torque has only a $z$ component.** As in Sec. 39.4, with $\boldsymbol{H} = H_0\cos\omega t\,\hat{\boldsymbol{x}}$ and $\boldsymbol{M} = (M_x, M_y, 0)$,

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -M_y H_x. \tag{65.2.1}$$

**Step 2: substitute the first-order magnetisation.** By (64.1.1) and (64.3.3), $M_y = M^{(0)}F\tau_B\Omega = \varphi M_s L(\xi_e)F\tau_B\Omega$, so

$$\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z = -\varphi M_s L\big(\xi_e\big)F\big(\xi_e\big)\tau_B\Omega\cdot H_0\cos\omega t. \tag{65.2.2}$$

**Step 3: average.** The constants $\varphi M_s$, $\tau_B$, $\Omega$, $H_0$ pass outside:

$$\overline{\big(\boldsymbol{M}\times\boldsymbol{H}\big)_z} = -\varphi M_s H_0\tau_B\Omega\,\overline{\cos\omega t\, L\big(\xi_e\big)F\big(\xi_e\big)}. \tag{65.2.3}$$

**Step 4: evaluate the prefactor.** Using $\varphi M_s = nm$ [Sec. 10.4], $\tau_B = 3\eta V/kT$ and $\varphi = nV$:

$$\varphi M_s H_0 \tau_B = n\,m\,H_0\,\frac{3\eta V}{kT} \tag{65.2.4}$$

$$= 3\eta\,(nV)\,\frac{mH_0}{kT} \tag{65.2.5}$$

$$= 3\eta\varphi\,\xi_0. \tag{65.2.6}$$

This is the same combination met at Eq. (52.4.4) of Volume VI. ✓

**Step 5: assemble.** Substituting (65.2.6) into (65.2.3), and writing the result as a vector along $\boldsymbol{\Omega} = \Omega\hat{\boldsymbol{z}}$:

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -3\eta\varphi\xi_0\,\overline{\cos\omega t\,L_eF}\;\boldsymbol{\Omega} = -6\eta\varphi\left[\frac{\xi_0}{2}\overline{\cos\omega t\,L_eF}\right]\boldsymbol{\Omega}, \tag{65.2.7}$$

so that with the definition

$$g \equiv \frac{\xi_0}{2}\,\overline{\cos\omega t\,L\big(\xi_e\big)F\big(\xi_e\big)} \tag{65.2.8}$$

we obtain

$$\overline{\boldsymbol{M}\times\boldsymbol{H}} = -6\eta\varphi\,g\,\boldsymbol{\Omega}, \tag{B-9}$$

■

## 65.3 The Particle Spin and the Viscosity

**The spin.** Substitute (B-9) into the torque balance (B-1), $6\eta\varphi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \overline{\boldsymbol{M}\times\boldsymbol{H}}$:

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = -6\eta\varphi g\boldsymbol{\Omega} \tag{65.3.1}$$

$$\boldsymbol{\omega}_p - \boldsymbol{\Omega} = -g\boldsymbol{\Omega} \tag{65.3.2}$$

$$\boxed{\boldsymbol{\omega}_p = \big(1-g\big)\boldsymbol{\Omega}} \tag{65.3.3}$$

exactly as the paper states.

**The viscosity.** From (65.3.3), $(\Omega - \omega_p)/\Omega = g$, and (B-2) gives

$$\boxed{\Delta\eta = \frac{3}{2}\,\eta\varphi\,g} \tag{65.3.4}$$

So $g$ is nothing but the rotational viscosity in units of its saturation value:

$$g = \frac{\Delta\eta}{\frac{3}{2}\eta\varphi} = \frac{\Delta\eta\big(\xi_0,\omega\tau_B\big)}{\Delta\eta(\infty,0)} = \frac{\eta_r\big(\xi_0,\omega\tau_B\big)}{\eta_r(\infty,0)}, \tag{65.3.5}$$

which is why the paper plots $g$ in Figs. 3 and 4 and compares it directly with the measured reduced viscosities of Fig. 5.

## 65.4 Physical Interpretation

Everything has again come down to a correlation. The number $g$ measures how much of the transverse response $L_eF$ is *in phase* with the driving field $\cos\omega t$. Positive correlation means the grains lag, the torque brakes them, and the viscosity rises; negative correlation means the grains are driven ahead of the flow and the viscosity falls. The structure is identical to $\overline{G\gamma}$ in Volume VI and to $\cos 2\alpha$ in Volume V. Three different formalisms, one mechanism.

## 65.5 Connection to the Next Equation

Substituting the averaged torque into the momentum equation gives (B-10) and closes the theoretical part of the paper.

# Chapter 66

# Equation (B-10): The Final Grouping

$$\eta\Delta\boldsymbol{v} - 3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega} = -2\eta\left(1+\frac{3}{2}\varphi g\right)\operatorname{curl}\boldsymbol{\Omega}. \tag{B-10}$$

## 66.1 Step-by-Step Derivation

**Step 1: the magnetic force density.** By (B-3), the magnetic term of the averaged momentum equation is $\frac{1}{2}\operatorname{curl}\overline{\boldsymbol{M}\times\boldsymbol{H}}$. With (B-9),

$$\tfrac{1}{2}\operatorname{curl}\overline{\boldsymbol{M}\times\boldsymbol{H}} = \tfrac{1}{2}\operatorname{curl}\big(-6\eta\varphi g\boldsymbol{\Omega}\big) = -3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega}, \tag{66.1.1}$$

which is the second term on the left of (B-10).

**Step 2: the viscous term.** By Eq. (41.2.6),

$$\eta\Delta\boldsymbol{v} = -2\eta\,\operatorname{curl}\boldsymbol{\Omega}. \tag{66.1.2}$$

**Step 3: add.**

$$\begin{aligned}\eta\Delta\boldsymbol{v} - 3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega} &= -2\eta\,\operatorname{curl}\boldsymbol{\Omega} - 3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega} &(66.1.3)\\ &= -2\eta\left(1+\frac{3\varphi g}{2}\right)\operatorname{curl}\boldsymbol{\Omega}, &(B\text{-}10)\end{aligned}$$

having extracted the factor $-2\eta$: $3\eta\varphi g = 2\eta\cdot\frac{3}{2}\varphi g$. ■

## 66.2 Reading the Result

The bracket $\left(1+\frac{3}{2}\varphi g\right)$ is the viscosity in units of $\eta$, so

$$\eta_{\text{total}} = \eta\left(1+\frac{3}{2}\varphi g\right), \qquad \Delta\eta = \frac{3}{2}\eta\varphi g, \tag{66.2.1}$$

in agreement with (65.3.4). Everything the ferrofluid does, in this theory, is contained in the single number $g(\xi_0,\omega\tau_B)$.

# Chapter 67

# The Numerical Procedure and the Theoretical Figures

## 67.1 The computation

The paper describes the procedure: for arbitrary values of the field amplitude and frequency the problem is solved numerically, by integrating the two differential equations (B-7) and (B-8) and then computing the integral over the period in (B-9). For prescribed $\xi_0$ and $\omega\tau_B$ the equations are solved by the fourth-order Runge–Kutta method with the initial conditions $\xi_e = \xi_0$ and $F \simeq 0$ at $t \simeq 0$.

Written out, the algorithm is:

1. Choose $\xi_0$ and $\omega\tau_B$. Set $t = 0$, $\xi_e = \xi_0$, $F = 0$.

2. Advance (B-7) and (B-8) together by fourth-order Runge–Kutta. They are coupled one way only — (B-7) does not contain $F$ — so $\xi_e(t)$ could in principle be found first, but it is simplest to step both at once.

3. Integrate for several periods until the transients die and the solution is periodic. (The initial conditions are not the periodic ones; the choice $\xi_e = \xi_0$, $F = 0$ is merely convenient, and the attractor is unique.)

4. Over one final period, accumulate the average $\overline{\cos\omega t\, L(\xi_e) F}$ and form $g = \frac{1}{2}\xi_0 \times$ that average, by (65.2.8).

5. Repeat on a grid of $(\xi_0, \omega\tau_B)$.

**Why $\xi_e$ is well behaved.** One might worry about (B-7) when $\cos\omega t$ passes through zero, since $\xi_0 \cos\omega t/\xi_e$ then vanishes and $\xi_e$ is driven towards zero, where $(\mathrm{d}\ln L_e/\mathrm{d}\xi_e)^{-1} \to \xi_e \to 0$. In fact both the prefactor and the driving vanish together and the equation remains regular: near $\xi_e = 0$ it reduces to $\tau_B\dot{\xi}_e = -(\xi_e - \xi_0\cos\omega t)$, a perfectly ordinary linear equation which carries $\xi_e$ smoothly through zero into negative values (the effective field simply reverses with the applied one).

## 67.2 Figure 3 of Bacri *et al.*: $g$ versus field

The figure plots

$$g\big(\xi_0, \omega\tau_B\big) = \frac{\eta_r\big(\xi_0, \omega\tau_B\big)}{\eta_r(\infty, 0)} \tag{67.2.1}$$

against $\xi_0$ (from 0 to 20), for $\omega\tau_B$ between 0.1 and 2. The qualitative content, now fully explicable:

- For $\omega\tau_B = 0.1$ the curve rises monotonically towards $\approx 0.8$: essentially the stationary behaviour of (SM-2), slightly reduced.
- For $\omega\tau_B = 1$ the curve is nearly flat and small: this is the crossover, where the in-phase correlation in (65.2.8) nearly vanishes.
- For $\omega\tau_B = 1.5$ and 2 the curves go *negative*, dip to a minimum near $\xi_0 \approx 6$–$8$, and then rise back towards zero as $\xi_0$ grows further.

**The non-monotonicity, explained.** At large $\xi_0$ the effective field is stiff — Sec. 63.5 showed the restoring rate grows as $\xi_e^2$ — so $\xi_e$ tracks the applied field ever more closely, the lag shrinks, and the correlation weakens. Hence $|g|$ falls again at very large fields. This is the same competition identified in Sec. 57.3 for the Shliomis–Morozov theory, but with the crucial difference that here the stiffening is computed from the Fokker–Planck-derived relaxation rather than assumed.

**The discrepancy that motivated the whole exercise.** Bacri *et al.* point out that with the phenomenological equation the computed $\Delta\eta(\xi)$ has a maximum at $\omega\tau_B < 1$ and no minimum at $\omega\tau_B > 1$, whereas the measurements show the opposite. Figure 3, computed from the effective-field theory, does show a minimum at $\omega\tau_B > 1$, in agreement with the data. The improvement is entirely due to the anisotropic, field-dependent relaxation of Sec. 62.7.

## 67.3 Figure 4: the theoretical isolines

More convenient than a family of curves is a contour map: isolines $g = \text{const}$ in the plane $(\xi_0, \omega\tau_B)$, for $g$ between $+0.8$ and $-0.9$. The isoline $g = 0$ divides the plane into two regions,

$$g > 0 \text{ for } \omega\tau_B \lesssim 1, \qquad g < 0 \text{ for } \omega\tau_B \gtrsim 1, \tag{67.3.1}$$

so that the sign change occurs at $\omega\tau_B \sim 1$ almost independently of the field amplitude — a robust prediction, and the one that experiment can most cleanly test.

# Chapter 68

# The Experiment

## 68.1 The fluid

The magnetic fluid is prepared by Massart's method; the colloidal particles are cobalt ferrite with anisotropy constant $K = 2 \times 10^6$ erg cm$^{-3}$, grain magnetisation $M_s = 400$ G and typical size 10 nm. The large anisotropy freezes the moment into the crystal up to fields of order $10^4$ Oe — the rigid-dipole condition, which we verified quantitatively in Sec. 17.6 by computing $\tau_N \approx 100$ s. To maximise the effect the suspension is concentrated: volume fraction $\varphi = 20\%$ in water.

**The design logic, in one line.** Every one of these choices follows from the theory. High $M_s$ and high $\varphi$ maximise $\Delta\eta = \frac{3}{2}\eta\varphi g$; high $K$ guarantees rigid dipoles so that $g \neq 0$ at all; and the particle size places $\tau_B$ such that $\omega\tau_B \sim 1$ falls in the audio range, where a solenoid can actually be driven.

## 68.2 The apparatus and the measurement

A Poiseuille flow in a horizontal capillary of diameter 1 mm is used for the viscosity measurement. The pressure gradient is provided by the difference $h$ in ferrofluid level between inlet and outlet, and the mass flow rate $Q$ is weighed at the outlet. The tube sits inside a solenoid producing a field along the flow direction up to $H = 1$ kOe at frequencies $f = \omega/2\pi$ from 0 to 1 kHz.

**Why this geometry.** As shown in Sec. 3.5, a cylindrical Poiseuille flow has azimuthal vorticity, while the solenoid field is axial. Hence $\boldsymbol{H} \perp \boldsymbol{\Omega}$, i.e. $\beta = \pi/2$, which by (SM-2) maximises the effect. It also means the demagnetising field vanishes (Sec. 22.6), so the internal field equals the applied one.

**Figure 1: the Newtonian check.** $Q$ is measured as a function of $h$ for $H = 0$ and $H = 2000$ Oe. Both data sets fall on straight lines through the origin, which establishes that the fluid is Newtonian with and without field, and the slopes give the viscosities. This is a necessary check: the entire theory computes a *viscosity*, i.e. assumes a linear relation between stress and shear rate, and Sec. 15.5, Step 12 showed that the theory does predict $\Omega$ to cancel out of $\Delta\eta$. Figure 1 confirms it experimentally. ✓

## 68.3 The numbers

| Quantity | Value | Source |
|---|---|---|
| solvent (water) viscosity | 1 cP | — |
| $\eta(0,0)$, zero field | 77 cP | Fig. 1 slope |
| $\eta(2000\ \text{Oe}, 0)$, static | 220 cP | Fig. 1 slope |
| $\Delta\eta$ at $f=0$, $H=2000$ Oe | $220-77=143$ cP | |
| $\eta$ at $H\sim 2000$ Oe, $f=50$ Hz | 100 cP | |
| $\eta$ at $H\sim 2000$ Oe, $f=250$ Hz | 77 cP | |
| $\eta$ at $H\sim 2000$ Oe, $f=700$ Hz | 58 cP | |
| $\Delta\eta$ at $f\geq 700$ Hz | $58-77=-19$ cP | |
| $\eta_r(\infty,0)$ | $\simeq 2.5$ | Fig. 2 asymptote |

**The central experimental result.** At $f=250$ Hz the viscosity in a 2000 Oe field is 77 cP — *exactly the zero-field value.* The magnetic field, at that frequency, does nothing at all to the viscosity: $\Delta\eta=0$. Above that frequency it *reduces* the viscosity below its zero-field value, reaching 58 cP at 700 Hz. The reduction is

$$\frac{77-58}{77}=0.247\approx 25\%, \tag{68.3.1}$$

the figure quoted in the paper. *This is the measurement of a negative rotational viscosity*, and it is the first such measurement reported.

**Consistency with the theory's frequency scale.** The sign change is observed near 250 Hz. With the calibration $\omega\tau_B = 10^{-2} f(\text{Hz})$ from Sec. 17.5, that is $\omega\tau_B \approx 2.5$. The theory of Sec. 67.3 predicts the change of sign near $\omega\tau_B\sim 1$. The agreement is of the right order but not exact — unsurprising given the polydispersity of a real ferrofluid, which smears $\tau_B$ over a range, and the concentration $\varphi=0.2$, at which interparticle interactions are certainly not negligible.

## 68.4 Figure 2: the raw data

The reduced viscosity

$$\eta_r(H,f)=\frac{\Delta\eta}{\eta(0,0)}=\frac{\eta(H,f)-\eta(0,0)}{\eta(0,0)} \tag{68.4.1}$$

is plotted against $H$ for $f=0,52,150,345,645,1480$ Hz. The behaviour:

- At $f=0$ the curve rises monotonically towards $\eta_r(\infty,0)\simeq 2.5$ — this is the McTague effect, first measured in 1969 and explained by Shliomis in 1972, i.e. our Eq. (SM-2).
- At 52 Hz the shape is essentially unchanged.
- At 150 Hz the shape changes qualitatively: $\eta_r$ first decreases, becomes negative, passes through a minimum, and only then rises towards its asymptote.
- At higher frequencies the same happens, with a deeper minimum.

**The non-monotonic $\eta_r(H)$.** This is precisely the behaviour predicted by the effective-field theory in Fig. 3 (Sec. 67.2) and *not* by the phenomenological one. It is the experimental signature of the field-dependent relaxation of Sec. 62.7.

## 68.5 Figure 5: the experimental isolines, and the comparison

For a quantitative comparison the data are replotted as isolines of $g$ in the plane $(\xi_0, \omega\tau_B)$, using

$$\eta_r(\infty, 0) = 2.5, \qquad \xi_0 = \frac{M_s V H}{kT} = 5.25 \times 10^{-3} H \text{ (Oe)}, \qquad \omega\tau_B = 2\pi f \tau_B = 10^{-2} f \text{ (Hz)}. \tag{68.5.1}$$

Both calibrations were verified from first principles in earlier volumes: the first in Sec. 23.5, where $m = M_s V$ with $M_s = 400$ G and $d \simeq 10$ nm gives $5.1 \times 10^{-3}$ and back-solving the paper's figure gives $d = 10.1$ nm; the second in Sec. 17.5, where $2\pi \times 1.6 \times 10^{-3} = 1.005 \times 10^{-2}$. ✓

The experimental isolines are drawn for the values

$$g = 0.216, \quad 0.1, \quad 0.044, \quad 0, \quad -0.036, \quad -0.052, \quad -0.068, \quad -0.092.$$

Comparing Fig. 5 with the theoretical Fig. 4, the authors report a strong similarity, and in particular that in both cases $g$ changes sign at $\omega\tau_B \sim 1$.

**What the comparison does and does not establish.** It establishes that a single-particle theory — no interactions, monodisperse, rigid dipoles — captures the *topology* of the response in the $(\xi_0, \omega\tau_B)$ plane: where the sign changes, where the extrema lie, how the curves bend. It does not establish quantitative agreement in magnitude, and the authors do not claim it: the measured isolines reach only $g \approx -0.09$ whereas the theory permits $g$ down to $-0.9$, and the measured $\eta_r(\infty, 0) = 2.5$ exceeds the dilute prediction $\frac{3}{2}\varphi = 0.3$ by a factor of eight (Sec. 15.6). At $\varphi = 0.2$ that is expected.

## 68.6 Conclusions of the Experimental Paper

**The result.** For $\omega\tau_B \gtrsim 1$ the viscosity of a ferrofluid in an alternating field can be *smaller* than in zero field. A decrease of 25% is observed.

**The limit.** An interesting limit would be to reach absolute zero viscosity, at which instabilities and flow recirculations would develop spontaneously without any external flow — the pump of Sec. 58.3. The marginal curve predicted by Shliomis & Morozov corresponds to $\xi_c = 126$ and $\omega_c\tau_B = 20$. Such values could be reached only with larger particles, for which both colloidal stability and Newtonian behaviour would be difficult to preserve. This is the same conclusion reached in Sec. 58.4, now stated from the experimental side: it is not the field that is out of reach but the particle.

**The application.** The result that survives is the tunability: the viscosity of this Newtonian magnetic fluid can be tuned between 220 and 50 cP by a field of 2000 Oe, monitoring *only* the frequency between 0 and 700 Hz. A factor of four in viscosity, controlled electronically, with no moving parts and no change of field amplitude. The authors suggest this could lead to a new generation of adaptive dampers — and that suggestion, thirty years on, remains the principal engineering legacy of these two papers.

# Chapter 69

# The Project Complete

## 69.1 What has been derived

Every equation in both papers has now been obtained from first principles, with no step omitted and no result assumed. The full index:

| Equation | Content | Volume, chapter |
|---|---|---|
| (SM-1) | $\tau_s = \rho_s d^2/60\eta$ | I, Ch. 7 |
| (SM-2) | stationary rotational viscosity | II, Ch. 15 |
| (SM-3), (SM-4) | weak- and strong-field limits | II, Ch. 16 |
| (SM-5) | $\tau_B = 3\eta V/kT$ | II, Ch. 17 |
| (SM-6) | fluid motion with spin coupling | III, Ch. 19 |
| (SM-7) | magnetisation relaxation | III, Ch. 20 |
| (SM-8) | rotational motion of the particles | III, Ch. 21 |
| (SM-9) | incompressibility and magnetostatics | III, Ch. 22 |
| (SM-10) | Langevin magnetisation | III, Ch. 23 |
| (SM-11) | inertialess spin | III, Ch. 24 |
| (SM-12) | reduced momentum equation | III, Ch. 25 |
| (SM-13) | reduced magnetisation equation | III, Ch. 26 |
| (SM-14) | the alternating field | III, Ch. 27 |
| (SM-15) | linear magnetisation law | III, Ch. 28 |
| (SM-16) | circular decomposition | IV, Ch. 30 |
| (SM-17) | rotating ansatz | IV, Ch. 31 |
| (SM-18) | the three relations | IV, Ch. 32 |
| (SM-19) | elimination of the phase | IV, Ch. 33 |
| (SM-20) | the vortex breaks the symmetry | IV, Ch. 34 |
| (SM-21) | rotational Doppler shift | IV, Ch. 35 |
| (SM-22) | transverse magnetisation of each component | IV, Ch. 36 |
| (SM-23), (SM-24) | $M_y$ and $\tan\alpha = \omega\tau_B$ | IV, Ch. 37 |
| (SM-25) | instantaneous torque | V, Ch. 39 |
| (SM-26) | two-scale averaging | V, Ch. 40 |
| (SM-27) | $\Delta\eta = \frac{1}{4}\eta\varphi\xi^2\cos^2\alpha\cos 2\alpha$ | V, Ch. 41 |

| Equation | Content | Volume, chapter |
|---|---|---|
| (SM-28) | **the central result** | V, Ch. 42 |
| (SM-29) | high-frequency asymptote | V, Ch. 43 |
| (SM-30)–(SM-32) | the second variant | V, Chs. 44–46 |
| (SM-33) | mean particle spin *(misprint corrected)* | V, Ch. 47 |
| (SM-34) | the master formula | V, Ch. 48 |
| (SM-35) | nonlinear magnetisation equation | VI, Ch. 50 |
| (SM-36) | coupled $F, G$ equations | VI, Ch. 51 |
| (SM-37) | averaged torque at arbitrary amplitude | VI, Ch. 52 |
| (SM-38), (SM-39) | general viscosity formula | VI, Ch. 53 |
| (SM-40) | square-wave solution *(misprint corrected)* | VI, Ch. 54 |
| (SM-41) | half-period reduction | VI, Ch. 55 |
| (SM-42), (SM-43) | formal solution | VI, Ch. 56 |
| (B-1)–(B-4) | torque balance, $\Delta\eta$, momentum, magnetisation | II–III |
| (B-5) | effective-field ansatz | VII, Ch. 61 |
| (B-6) | effective-field evolution *($L_3$ identified)* | VII, Ch. 62 |
| (B-7) | zero approximation | VII, Ch. 63 |
| (B-8) | first-order response | VII, Ch. 64 |
| (B-9) | mean torque and $g$ | VII, Ch. 65 |
| (B-10) | final grouping | VII, Ch. 66 |

## 69.2 Three editorial findings

In the course of the derivations three points emerged where the printed papers and the mathematics disagree. All three were resolved in favour of the mathematics, and in each case the papers' own numbers confirm the correction.

1. **(SM-33)** is printed with $1 - \xi^2\tau_B^2$ in the numerator. This is dimensionally impossible ($\xi$ is a pure number) and inconsistent with (SM-28) and (SM-34); it should read $1 - \omega^2\tau_B^2$. [Volume V, Sec. 47.1.]

2. **(SM-40)** is printed with $\tanh\varepsilon$ where the derivation gives $\tanh^2\varepsilon$. The corrected form reproduces all three of the authors' own quoted values — $\omega_0\tau_B = 0.98$, $\omega_m\tau_B = 1.707$, $\Delta\eta_m = -0.105\Delta\eta(0)$ — and the printed form does not. [Volume VI, Secs. 54.1 and 54.6.]

3. **(B-6)** contains an undefined symbol $L_3$. Matching against the exact moment equation with the effective-field closure identifies it as $L_3 = L(\xi_e)$, and the independent derivation of (B-8) confirms it. [Volume VII, Secs. 62.5 and 64.5.]

## 69.3 The physics, in one page

If everything else were forgotten, this is what should remain.

A ferrofluid is a suspension of magnetic grains small enough to be kept apart by thermal motion. Each grain can *spin* independently of the liquid around it, and that single extra degree of freedom is the whole subject.

In a shear flow the liquid rotates locally at $\boldsymbol{\Omega} = \tfrac{1}{2}\nabla \times \boldsymbol{v}$, and without a field the grains simply turn with it: nothing happens. Apply a stationary field and the grains' moments are held; the grains cannot turn freely; the liquid must shear past them; the viscosity rises by $\Delta\eta = \tfrac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$, which saturates at $\tfrac{3}{2}\eta\varphi$ when the grains are locked. A permanent magnet does no work, so thermodynamics guarantees this is always an increase.

Apply instead an *alternating* field along a fixed axis. It picks out an axis but no sense of rotation, so in a fluid at rest half the grains swing one way and half the other and the mean spin is exactly zero. Introduce a shear, and the degeneracy breaks: a grain carried round by the flow sees the two circular components of the field at Doppler-shifted frequencies $\omega \mp \Omega$, responds differently to each, and acquires a net transverse magnetisation proportional to $\Omega$ and lagging the field by *twice* the single-component phase lag. Crossed with the field, that magnetisation gives an average torque proportional to $-\Omega \cos 2\alpha$ — and $\cos 2\alpha$ reverses sign when $\alpha$ passes $\pi/4$, that is when $\omega\tau_B = 1$.

Above that frequency the torque no longer brakes the grains: it drives them, they spin *faster* than the liquid, and they feed angular momentum into the flow. The rotational viscosity becomes negative. There is no paradox — the generator driving the solenoid is doing the work, and part of it arrives as kinetic energy of the fluid.

The effect is real, it was measured, and it is modest: a 25% reduction below the zero-field viscosity. Driving the *total* viscosity to zero, which would turn the fluid into a self-driven pump, would require grains so large that they would neither stay suspended nor keep their moments frozen. What remains, and what may yet be useful, is a liquid whose viscosity can be changed fourfold by turning a frequency dial.

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*End of the work.*

Volumes I–VII, Chapters 1–69, pages 1–239.